QP CODE**:**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

FOURTH SEMESTER

Discipline Specific Elective (DSE) – MG4DSEMBG200 –

INDUSTRIAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** mark

1. Name one organic acid produced by microbes. [K] / [CO1]
2. Comment on monosodium glutamate. [U] / [CO4]
3. Identify any one product concentration method. [K] / [CO3]
4. What are filter aids? [U] / [CO3]
5. Mention any one role of microbes in the food industry. [K] / [CO1]

[1x5 = 5]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

6. What was the contribution of Louis Pasteur to industrial microbiology? [U] / [CO1]
7. What are precursors? give examples. [U] / [CO1]

8.Give any two examples of Primary screening.	[K] / [CO1]
9.What is Dual fermentation?	[U] / [CO2]
10. What is a chemostat?	[K] / [CO2]
11. What is lyophilization?	[K] / [CO1]
12. Explain the danger Spots in Fermenters	[U] / [CO2]
13. Mention any two advantages of SSF.	[K] / [CO2]
14. Name any two methods of air sterilization.	[K] / [CO2]
15. Why moist heat is preferred over dry heat for fermenter sterilization?	[U] / [CO2]
16. Name any two materials used as supports for immobilization.	[K] / [CO3]
17. Define mechanical cell disruption and give one example.	[U] / [CO3]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18.Analyze how different product formulations affect the stability and effectiveness of microbial products.	[An] [CO3]
19. Explain in detail the characteristics of industrial strain and its importance	[U/[CO1]
20. Analyze the steps involved in downstream processing	[An] / [CO3]
21. Explain in detail the commercial importance of the enzyme Amylase and its production.	[U] / [CO4]
22. Describe the structural components of a fermenter involved in aeration and agitation.	[U] / [CO2]
23. Explain the production process of amylase and its commercial importance.	[U] / [CO4]

24. Assess the benefits and drawbacks of using immobilized enzymes and decide if they are better than free enzymes. [E] / [CO3]

[6x5 = 30]

Part D

Long Essay Questions

Answer any **1** question

Each question carries **15** marks

25. Explain fermentation. Write in detail about any two types of fermentation. [U] / [CO2]

26. What are antibiotics? Explain in detail the production of penicillin. [U] / [CO4]

[15x1 = 15]